

MA 311: Mathematical Modeling

Course Code	MA 311
Course Name	Mathematical Modeling
L-T-P-C	3-0-0-3
Prerequisites	Ordinary and Partial Differential Equations, Linear Algebra, Basics of Programming, Basic of Statistics
Intended for	B.Tech. Mathematics & Computing
Distribution	Core for BTech. Mathematics and Computing, Elective for other branches
Mutual Exclusion	-
Approval	59th BoA (Apr 2025)

Course Contents

- **Introduction to Mathematical Modeling:** Modeling Process, Types of Models, Applications in Science and Engineering, Modeling Change, Model Fitting Techniques, Experimental Modeling, Simulation Modeling. [9 Lectures]
- **Modeling with System of Differential Equations I:** Compartmental Models, Population Models: Single Species Models (Exponential, Logistic, and Gompertz growth), Interacting Population Models: Prey-Predator Models. [9 Lectures]
- **Modeling with System of Differential Equations II:** Chemostat Models, Competition Model, Mutualism Models, Tumor Models, Structured Population Models, spatially structured and Age structured Models, Cooperative Model, Population Dispersal Model based on Diffusion, Traveling Wave Solutions, Fisher's Equation. [11 Lectures]
- **Mathematical Modeling with Machine Learning:** Neural Network Modeling, Neural ODEs, Physics-Informed Neural Networks (PINNs), The Pareto Front, Model Selection criteria, Cross-Validation, Information Criteria. [7 Lectures]
- **Case Study:** Oscillator Generated Wave Phenomena and Central Pattern Generators, The spruce budworm outbreak model, SEIR model, PINNs for solving Heat Equation. [5 Lectures]

Text Books

1. Leah Edelstein Keshet, **Mathematical Models in Biology**, SIAM, 2005.
2. Mark Kot, **Elements of Mathematical Ecology**, Cambridge University Press, 2008.
3. Strogatz, S. H., **Nonlinear dynamics and chaos: With applications to physics, biology, chemistry, and engineering**, 2nd Edition, Westview Press, 2015.
4. Haykin, Simon., **Neural Networks: A Comprehensive Foundation**, 2nd Edition, Prentice Hall, 1999.

References

1. C.M. Bishop, **Pattern Recognition and Machine Learning**, Springer, 2006.
2. Strang, Gilbert, **Linear algebra and learning from data**, Cambridge: Wellesley-Cambridge Press, 2019.
3. C. Eck, H. Garcke and P. Knabner, **Mathematical Modeling**, Springer publication, 2017.
4. R. Temam, A.M. Miranville, **Mathematical Modelling in Continuum Mechanics**, Cambridge University Press, Cambridge, 2001.
5. Friedman, Avner, and Chiu-Yen Chou., **Introduction to Mathematical Biology**, Springer, 2018.
6. Giordano, Frank R., William P. Fox, and Steven B. Horton., **A first course in mathematical modeling**, Cengage Learning, 2013.